



Intelligent phishing website detection: A CNN-SVM approach with nature-inspired hyperparameter tuning

Santosh Kumar Birthriya*, Priyanka Ahlawat, Ankit Kumar Jain

Department of Computer Engineering, National Institute of Technology Kurukshetra, Haryana, India

ARTICLE INFO

Keywords:

Phishing
Website
Support vector machine
Convolutional neural networks
Swarm intelligence
Cybersecurity

ABSTRACT

Phishing attacks represent a growing threat to online users and software developers, necessitating the development of advanced detection strategies. This study proposes a hybrid framework that integrates convolutional neural networks (CNN) for feature extraction and support vector machines (SVM) for classification, with the SVM optimized using the grey wolf optimizer (GWO). The CNN component is responsible for capturing complex and discriminative patterns from website data, enabling more effective differentiation between phishing and legitimate websites. Hyperparameter tuning via GWO enhances the classification performance of the SVM by generating an optimal decision boundary. Evaluation was conducted using established datasets, including those from Kaggle, the UCI Machine Learning Repository, Phishtank, 5000 Best Websites, and Alexa. Experimental results show that the CNN-SVM model, with GWO optimization, achieved an accuracy of 99.18 %, indicating its practical utility in phishing detection applications. The findings suggest that the proposed framework, supported by additional security mechanisms, contributes to a reduction in false positives while maintaining reliable detection of phishing threats.

1. Introduction

Phishing has emerged as a significant threat in the context of expanding e-commerce and online transactions, making it imperative to safeguard digital resources [1]. Phishing attacks commonly deceive users by directing them to fraudulent websites that closely resemble legitimate platforms, thereby eliciting sensitive information such as credit card details, passwords, and other personal data [2]. This threat persists on a global scale due to the continuous proliferation of online activities, wherein attackers exploit the reputation of trusted organizations to extract protected information, including login credentials and payment details [3]. Phishing schemes can employ multiple ways to trick users into giving out sensitive information, these may range from false advertisement, voice mails, texts, social network interactions or even infecting the user's devices with spyware [4].

The Fig. 1 depicts the quarterly trends in the total number of reported unique phishing websites within the time range of Q1 2020-Q3 2023. There has been a noticeable increase in the number of phishing websites through time. Beginning with 164,772 in Q1 2020, it was observed that the numbers have risen with a culmination of reaching a peak of 1,624,144 in Q1 2023. Also, the information shows that the increase is also shown in the number of phishing websites with a big jump of the same in 2023, implying that phishing activi-

ties and attacks have been on the rise and more persistent than before [5].

Despite significant advancements in anti-spam measures that filter out many malicious attachments, phishing attacks continue to pose a threat due to the increasingly sophisticated tactics employed by attackers [6]. A classifier that integrates various indicators of phishing could greatly enhance fraud detection and prevention efforts [7,8]. Traditional GWO-SVM approaches that rely on manually curated features [9], the proposed framework employs a convolutional neural network (CNN) to automate feature extraction.

Most current approaches fail to adapt to rapidly shifting tactics, emphasizing the need for advanced, robust, and scalable methods. This study addresses three core questions:

- **RQ1:** How can high-level, meaningful features be automatically extracted from websites, capturing both structural and content-based characteristics indicative of phishing, while reducing the need for manual intervention in feature engineering?
- **RQ2:** How can the GWO be utilized to fine-tune the hyperparameters of the SVM classifier, thereby improving the detection accuracy and minimizing false positives in phishing detection systems?
- **RQ3:** To what extent can the hybrid CNN-SVM model, optimized by GWO, generalize across diverse and large-scale phishing datasets,

Peer review under responsibility of KeAi Communications Co., Ltd.

* Corresponding author.

E-mail addresses: cs.santosh1b@gmail.com (S.K. Birthriya), priyankaahlawat@nitkkr.ac.in (P. Ahlawat), ankitjain@nitkkr.ac.in (A.K. Jain).

<https://doi.org/10.1016/j.csa.2025.100100>

Received 13 November 2024; Received in revised form 1 May 2025; Accepted 28 May 2025

Available online 28 May 2025

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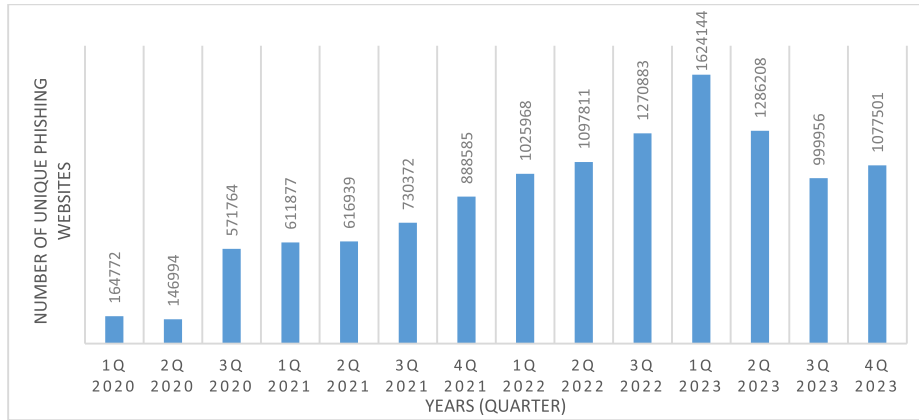


Fig. 1. Statistics of phishing websites 1Q 2020 – Q2023.

ensuring robust performance and scalability in real-world applications?

Problem Statement: Phishing attacks evolve rapidly, posing severe risks to online users and organizations. Existing detection systems often rely on manual feature engineering and struggle to adapt to emerging tactics. This necessitates a more automated, scalable, and accurate detection solution that minimizes false positives across diverse, large-scale datasets.

Contributions: This study proposes a novel phishing detection approach combining CNN and GWO to optimize SVM hyperparameters, improving feature extraction and detection accuracy.

- We propose a comprehensive preprocessing pipeline that applies iterative imputation, quantile-based normalization, and data shuffling to address missing values, normalize feature scales, and reduce bias in phishing datasets.
- Unlike previous GWO-SVM methods reliant on manual feature selection, our approach employs a CNN to learn complex, high-level representations directly from raw data.
- A CNN is employed for feature extraction to capture complex patterns, while an SVM is utilized for classification, enabling precise decision boundary formation to distinguish phishing websites from legitimate ones.
- The GWO is integrated to automatically determine optimal SVM hyperparameters (C and γ), reducing manual effort and enhancing detection accuracy.
- Experiments on multiple datasets (UCI, Kaggle, Phishtank, 5000 Best Websites, Alexa) demonstrate that the CNN–GWO–SVM framework delivers near-perfect accuracy, minimal false positives, and superior performance relative to existing solutions.

The structure of the paper is organized as follows. Section 2 reviews related work on phishing website detection. Section 3 discusses nature-inspired algorithms relevant to the study. Section 4 outlines the research methodology. Section 5 presents the experimental results and performance analysis. Section 6 reports the key findings and their significance. Section 7 concludes the study and highlights directions for future research.

2. Related Works

Phishing website detection algorithms typically utilize supervised machine learning techniques, mapping inputs to outputs through specific functions. During this process, classifiers identify features that help predict the outcome, determining if a website is legitimate or phishing.

Awasthi A et al. (2019) [9], a phishing classification system based on URL features and point-wise mutual information achieved 90.363 % accuracy with RF and an overall accuracy of 92.5 %. Orunsolu et al. (2022) [10], a model using URL, webpage properties, and behavior features, trained with SVM and NB, achieved 99.96 % accuracy but was

limited by a small dataset. Finally, Almomani A et al. (2022) [11] emphasized semantic features from URLs and domain characteristics, with gradient boosting classifier and random forest classifier achieving about 97 % accuracy across two datasets, while GaussianNB and SGD were the least accurate at 84 % and 81 %, respectively. Online tools often employ blacklisting to identify phishing sites [12]. One approach involves comparing the visual profiles of phishing sites with those of the legitimate sites they mimic and other similar phishing sites [13]. The PhishZoo method combines whitelisting and blacklisting by comparing sites against profiles of trusted websites to detect mismatches and identify phishing attempts [14]. Another strategy extracts and compares features from the text, visual elements, and overall appearance of websites, generating a signature to compute similarity scores and assess website legitimacy [15]. Additionally, some methods analyze web pages at multiple levels block, layout, and style to calculate a comprehensive similarity score for phishing detection [16]. Another technique converts web pages into low-resolution images and uses color and spatial features to create image signatures. The earth mover's distance (EMD) is then applied to these signatures to differentiate between phishing and normal web pages [17].

Fuzzy logic combined with neural networks are effectively classified phishing websites [18], while a fusion of fuzzy logic and data mining has targeted e-banking phishing detection [19]. Genetic Algorithms (GA) have been used to develop rules to prevent phishing attacks [20]. Simpler models, such as a single-hidden layer feedforward neural network with two neurons, have also been employed for this task [21]. Other techniques, including SVMs, Naive Bayes Classifiers, K-Means, K-Nearest Neighbors, and Affinity Propagation, have been utilized to detect malicious websites [22]. Specifically, SVMs achieved 84 % accuracy by analyzing a mix of semantic features and other characteristics in a dataset of 279 phishing and 100 legitimate websites [23]. URL-based approach for phishing detection used classifiers like Naive Bayes, Iterative Dichotomiser 3 (ID3), K-Nearest Neighbor (KNN), Decision Tree, and Random Forest. Incorporating Genetic Algorithms (GA) for feature selection significantly enhanced detection accuracy. Experimental results showed that combining ID3 with Yet Another Generating Genetic Algorithm (YAGGA) improved detection accuracy to 95 % [24].

Recent advancements in phishing website detection have leveraged various machine learning and optimization techniques to enhance accuracy and efficiency. A PSO-based feature weighting method [25] significantly improved classification accuracy to 96.83 %, optimizing true positive and negative rates while reducing false rates with fewer features. SVM classifier optimized with nature-inspired algorithms, notably the GWO, for analyzing URL properties, demonstrating superior performance [26]. An enhanced Binary Bat Algorithm [27] integrated with deep learning and the Adam optimizer achieved a 94.8 % accuracy in URL classification. The MOE/RF model [28], combining multi-objective evolution optimization and random forest, reached 99.04 % accuracy by minimizing false positives using innovative initialization and selection

Table 1
Comparative analysis for related works.

References	Optimization Method	SVM Hyperparameter Tuning	Accuracy
Suleman et al. (2019) [24]	Genetic Algorithms	NO	95 %
Ali et al. (2020) [25]	PSO-based feature selection	NO	96.83 %
Anupam et al. (2021) [26]	SVM Optimized using BAT, FA, GWO, and WO	Yes	91.83 %
Zhu et al. (2022) [27]	Multiobjective Evolution Optimization	NO	99.04 %
Nordin et al. (2023) [28]	Butterfly Optimization Algorithm (BOA), Harmony Search (HS)	NO	98.80 %
Kumar et al. (2023) [29]	Binary Bat Algorithm	NO	94.8 %
Alsenani et al. (2023) [30]	PSO-based feature selection	NO	97.81 %
Our Proposed Work	CNN + SVM Hyperparameter Optimization using GWO	Yes	99.18 %

Algorithm 1

SVM Hyperparameter Optimization Using BA.

- 1 **Initialize:** Set up a population of bats $\{b_i\}_{i=1}^n$ with random positions x_i and velocities v_i . Define frequency f_i , loudness A_i , and pulse emission rate r_i for each bat.
- 2 **Define Objective Function:** Evaluate the classification accuracy of SVM with hyperparameters C and γ .
- 3 **Optimization Loop:** each iteration $t = 1$ to T_{max} each bat $i = 1$ to n Update frequency:
 $f_i = f_{min} + (f_{max} - f_{min}) \cdot \theta$,
 where θ is a random number in $[0, 1]$. Update velocity and position:
 $v_i' = v_i^{t-1} + (x_i^{t-1} - x_{best})f_i$,
 $x_i' = x_i^{t-1} + v_i'$,
 where x_{best} is the current global best position. Generate a local solution if a random number $\epsilon > r_i$:
 $x_i^{new} = x_{best} + \delta A_i$,
 where δ is a random number in $[-1, 1]$. Evaluate the fitness of x_i' . Update x_{best} if a better solution is found. Adjust loudness and pulse emission rate:
 $A_i^{t+1} = \alpha A_i^t$, $r_i^{t+1} = r_i^0 [1 - \exp(-\gamma t)]$,
 where α and γ are constants.
- 4 **Output:** Return the best solution x_{best} with optimal hyperparameters C and γ .

strategies. An integrated BOA with Harmony Search [29] balanced exploration and exploitation, achieving accuracies of 98.69 % and 98.80 % on two UCI datasets. Additionally, PSO-based feature selection [30] improved various performance metrics, achieving 97.81 % and 90.39 % accuracy on phishing datasets using artificial neural networks.

Our work combines CNN with SVM optimization using grey wolf optimizer (GWO), achieving near-perfect accuracy, and demonstrating significant improvements in phishing detection, as shown in Table 1.

3. SVM hyperparameter optimization using nature-inspired algorithms

Nature-inspired algorithms are utilized to optimize the hyperparameters of the SVM, significantly enhancing the model's classification performance. These algorithms mimic natural phenomena and behaviors to efficiently explore the search space and find optimal parameter settings. The specific algorithms employed in this study include the bat algorithm, firefly algorithm, whale optimization algorithm, grey wolf optimizer, and ant colony optimization. Each algorithm is detailed below with its respective optimization procedure.

3.1. Bat algorithm (BA)

The BA inspiration from the echolocation behavior of microbats. These bats emit ultrasonic pulses and analyze the returning echoes to navigate and locate prey. In the optimization context, the algorithm simulates this behavior by guiding virtual bats through the search space. Each bat adjusts its position, velocity, frequency, loudness, and pulse emission rate dynamically, utilizing echolocation feedback to identify

the optimal solution [31]. The following Algorithm 1 describes the process for optimizing the hyperparameters C and γ of an SVM model using the BA.

3.2. Firefly Algorithm (FA)

The FA is a metaheuristic optimization technique inspired by the flashing behavior of fireflies, where the intensity of their light serves as an indicator of their attractiveness. In this Algorithm 2, each firefly represents a potential solution within the search space. Fireflies are inherently attracted to others with higher light intensity, causing them to move toward brighter individuals. The algorithm is governed by two key components: the variation in light intensity and the formulation of attractiveness. These elements collectively guide the movement of fireflies, enabling the exploration and exploitation of the search space to converge toward optimal solutions [32].

3.3. Whale optimization algorithm (WOA)

The WOA draws inspiration from the bubble-net hunting strategy of humpback whales. Candidate solutions represent whales' encircling and spiral movements around prey. The algorithm operates through two primary mechanisms: exploitation (encircling prey) and exploration (searching for prey). A probabilistic approach determines the selection of these mechanisms, ensuring an effective balance between local and global search processes. Encircling helps refine solutions, while exploration prevents premature convergence. This nature-inspired optimization technique proves effective in solving complex optimization problems by simulating cooperative behavior observed in whale hunting strategies [33].

Algorithm 2

SVM Hyperparameter Optimization Using FA.

- 1 **Initialize:** Set up a population of fireflies $\{f_i\}_{i=1}^n$ with random positions x_i . Define light absorption coefficient γ .
- 2 **Define Objective Function:** Evaluate the classification accuracy of SVM with hyperparameters C and γ .
- 3 **Optimization Loop:** each iteration $t = 1$ to T_{max} each firefly $i = 1$ to n each firefly $j = 1$ to n firefly j is brighter than firefly i Move firefly i towards firefly j :
 $x_i = x_i + \beta_0 e^{-\gamma r_{ij}} (x_j - x_i) + \alpha e_i$,
 where β_0 is the attractiveness at $r = 0$, r_{ij} is the distance between fireflies i and j , α is the randomization parameter, and e_i is a vector of random numbers. Evaluate the fitness of x_i . Rank the fireflies based on their brightness.
- 4 **Output:** Return the best solution with optimal hyperparameters C and γ .

Algorithm 3

SVM Hyperparameter Optimization Using WOA.

-
- 1 **Initialize:** Set up a population of whales $\{w_i\}_{i=1}^n$ with random positions x_i .
 - 2 **Define Objective Function:** Evaluate the classification accuracy of SVM with hyperparameters C and γ .
 - 3 **Optimization Loop:** each iteration $t = 1$ to T_{max} each whale $i = 1$ to n Update parameters a , A , and C :
 $a = 2 - \frac{2t}{T_{max}}$, $A = 2a \cdot r_1 - a$, $C = 2r_2$,
 where r_1, r_2 are random numbers in $[0, 1]$. Generate a random number p in $[0, 1]$. $p < 0.5$ $|A| < 1$ Update position using exploitation (encircling prey):
 $D = |C \cdot x_{best} - x_i|$, $x_i = x_{best} - A \cdot D$,
 where x_{best} is the best solution found so far. Select a random whale x_{rand} . Update position using exploration (searching for prey):
 $D = |C \cdot x_{rand} - x_i|$, $x_i = x_{rand} - A \cdot D$.
 Update position using spiral mechanism:
 $D' = |x_{best} - x_i|$, $x_i = D' e^{bl} \cos(2\pi l) + x_{best}$,
 where b is a constant and l is a random number in $[-1, 1]$. Evaluate the fitness of x_i . Update x_{best} if a better solution is found.
 - 4 **Output:** Return the best solution x_{best} with optimal hyperparameters C and γ .
-

Algorithm 4

SVM Hyperparameter Optimization Using GWO.

-
- 1 **Initialization:**
 - Set the number of wolves N , maximum iterations $MaxIter$, and the bounds for hyperparameters C and γ .
 - Randomly initialize the positions of N wolves within the defined bounds.
 - Evaluate the fitness of each wolf using the objective function.
 - 2 **Leader Identification:** Identify the best three wolves as alpha (α), beta (β), and delta (δ) based on their fitness values.
 - 3 **Optimization Loop:** each iteration from 1 to $MaxIter$ each wolf i in 1 to N Update the position of wolf i based on the positions of α , β , and δ using the GWO update Equations:

$$\vec{D}_\alpha = |\vec{C}_1 \cdot \vec{X}_\alpha - \vec{X}_i|$$

$$\vec{D}_\beta = |\vec{C}_2 \cdot \vec{X}_\beta - \vec{X}_i|$$

$$\vec{D}_\delta = |\vec{C}_3 \cdot \vec{X}_\delta - \vec{X}_i|$$

$$\vec{X}_i^{(t+1)} = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3}$$
 where: $\vec{X}_1 = \vec{X}_\alpha - A_1 \cdot \vec{D}_\alpha$, $\vec{X}_2 = \vec{X}_\beta - A_2 \cdot \vec{D}_\beta$, $\vec{X}_3 = \vec{X}_\delta - A_3 \cdot \vec{D}_\delta$.
 Ensure the new positions are within the bounds of C and γ .
 Evaluate the new positions of all wolves using the objective function.
 Update the positions of α , β , and δ based on the new fitness values.
 - 4 **Result:** Return the best solution found by the alpha wolf α .
-

Algorithm 5

SVM Hyperparameter Optimization Using ACO.

-
- 1 **Initialize:** Set up initial pheromone levels τ_{ij} for all possible transitions between nodes representing hyperparameter values. Define heuristic information η_{ij} based on problem-specific knowledge.
 - 2 **Define Objective Function:** Evaluate the classification accuracy of SVM with hyperparameters C and γ .
 - 3 **Optimization Loop:** each iteration $t = 1$ to T_{max} each ant $k = 1$ to m Construct a solution by moving through the parameter space according to transition probabilities:

$$P_{ij}^k = \frac{[\tau_{ij}]^\alpha [\eta_{ij}]^\beta}{\sum_{(i,j) \in allowed} [\tau_{ij}]^\alpha [\eta_{ij}]^\beta}$$
 where α and β control the influence of pheromone and heuristic information. Evaluate the fitness of the constructed solution.
 Update pheromone levels:

$$\tau_{ij} = (1 - \rho) \tau_{ij} + \sum_{k=1}^m \Delta \tau_{ij}^k$$
 where ρ is the evaporation rate and $\Delta \tau_{ij}^k$ is the pheromone deposited by ant k . Apply pheromone evaporation to prevent premature convergence.
 - 4 **Output:** Return the best solution with optimal hyperparameters C and γ .
-

3.4. Grey wolf optimizer (GWO)

The GWO algorithm mimics the social hierarchy and hunting strategies of grey wolves. The population divides into four roles: alpha (α), beta (β), delta (δ), and omega (ω), where α leads the pack. The optimization process relies on cooperative searching, where the leading wolves guide the exploration and exploitation of the solution space. This approach ensures efficient convergence toward optimal solutions. Algorithm 4 details the SVM hyperparameter optimization process using GWO. The method leverages leadership dynamics and group coordination to improve search efficiency in complex optimization problems [34].

3.5. Ant colony optimization (ACO)

ACO is inspired by the foraging behavior of ants and their ability to find the shortest paths using pheromone trails. In this Algorithm 5, artificial ants construct solutions by moving through the parameter space, guided by pheromone levels that represent the desirability of

choices based on previous successful paths. The algorithm simulates the pheromone deposition and evaporation processes to balance exploration and exploitation. Each of these algorithms leverages natural strategies to optimize SVM hyperparameters and iteratively improve model performance until the best results are achieved. Simulating natural behaviors such as echolocation, bioluminescence, social hunting, and pheromone communication, these algorithms effectively explore the hyperparameter space and enhance the classification accuracy of the SVM model [35].

4. Proposed methodology

The proposed methodology, as depicted in Fig. 2, utilizes a CNN to extract deep features from the input data. These features are then utilized as inputs for the SVM model. This hybrid approach leverages the powerful feature extraction capabilities of CNN and the robust classification abilities of SVM. The incorporation of GWO for hyperparameter optimization further refines the SVM classifier, resulting in enhanced classification performance. The performance of the hybrid CNN-SVM

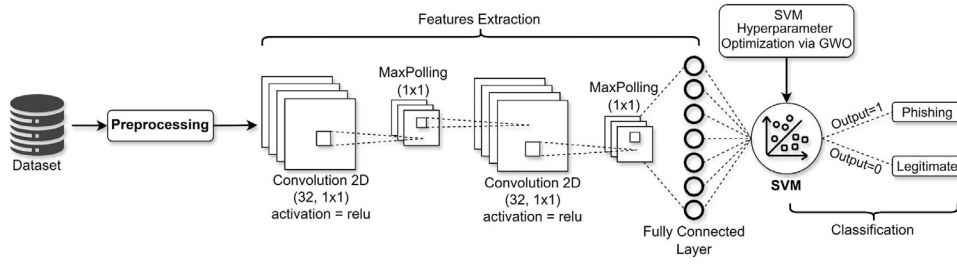


Fig. 2. Phishing website detection: a proposed architectural framework.

model is rigorously evaluated, demonstrating the substantial impact of GWO-based hyperparameter optimization on classifier efficacy.

4.1. Data preprocessing

The preprocessing stage is essential for ensuring the dataset is of high quality and adequately prepared for training machine learning models. This process involves addressing missing values, normalizing data, and shuffling the dataset to enhance model performance and reliability. Missing values are addressed using the Iterative Imputer from scikit-learn, which employs a predictive modeling approach to estimate missing entries by modeling each feature in relation to others, offering a more robust alternative to traditional methods such as mean or median imputation. Data normalization is performed using the QuantileTransformer, which transforms feature values into a normal distribution, ensuring consistent scaling and improving model stability during training. Lastly, the dataset is shuffled using the shuffle function from scikit-learn to eliminate ordering bias, thereby ensuring that the training and testing sets accurately represent the overall data distribution.

4.2. Feature extraction

The CNN architecture used to extract features from input data, focusing on identifying complex patterns that distinguish phishing websites from legitimate ones. The proposed CNN architecture consists of two convolutional layers, each followed by a max-pooling layer, and concludes with two fully connected layers. Specifically, each convolutional layer employs 32 kernels of size 1×1 with ReLU activation, allowing for channel-wise transformations while minimizing the number of parameters. Although 1×1 convolutions do not enlarge the spatial receptive field, they facilitate the learning of cross-channel interactions.

Following each convolutional layer, a 1×1 max-pooling operation is applied. Although the chosen pooling size does not significantly reduce the spatial dimensions, it ensures architectural consistency and adheres to typical CNN design patterns. The feature maps produced are subsequently flattened into a one-dimensional vector via a Flatten() operation.

Subsequently, the network employs a fully connected layer with 256 neurons (again with ReLU activation) to capture intermediate, high-level abstractions. This layer is followed by a softmax output layer designed to produce probability distributions over the target classes. During the training process, the network weights are updated via the Adam optimizer, with categorical cross-entropy serving as the loss function. Model checkpoints are saved based on validation accuracy to preserve only the most effective parameter configurations.

4.3. Classification and Optimization

SVM plays a critical role in the proposed phishing website detection system by acting as the final classifier. In this case, the two classes are phishing websites (labeled as 1) and legitimate websites (labeled as 0). The goal of the SVM is to maximize the margin between the two classes, which helps improve the model's generalization ability on unseen data. The features extracted by the CNN are passed to an SVM classifier, which

separates phishing from legitimate websites. The SVM constructs an optimal hyperplane that separates the two classes with maximum margin. If the classes are not linearly separable, SVM uses kernel functions to map the data into a higher-dimensional space, where separation becomes easier.

Optimization of SVM Hyperparameters using GWO: SVM achieve strong results across diverse domains, their performance critically depends on appropriate hyperparameter selection, namely the regularization parameter C and the kernel coefficient γ . This study employs the GWO to automate the search for the optimal combination of (C, γ) . The GWO is a population-based metaheuristic inspired by the social hierarchy and collaborative hunting behavior of grey wolves. The population is structured into:

- Alpha Wolves, which guide the overall optimization.
- Beta and Delta Wolves, which refine candidate solutions.
- Omega Wolves, which follow and adapt to leading wolves to ensure exploration of the search space.

The toggling between exploration (broad search) and exploitation (local refinement), the GWO converges to near-optimal or optimal solutions.

Optimization Process: The optimization begins by initializing a population of wolves in the (C, γ) parameter space. Each wolf corresponds to a candidate solution and is evaluated by training an SVM using the CNN-derived feature set, followed by measuring the classification accuracy on a validation subset. Two critical phases guide the wolves' movements:

- Exploration: Early iterations promote diversity, reducing the risk of suboptimal convergence by encouraging wolves to disperse throughout the parameter space.
- Exploitation: In later stages, wolves coalesce around the most promising solutions (alpha, beta, and delta positions), allowing fine-grained optimization of C and γ .

In optimization process a stopping criterion (e.g., maximum number of iterations) is met, the GWO outputs the best-performing hyperparameters. The approach proposed overcomes the weaknesses of completely end-to-end methods by dividing the responsibilities of feature learning (CNN) and decision boundary construction (SVM). Furthermore, the GWO-optimized hyperparameter adjustments very often improve performance during generalization. The results achieved prove that this combination of deep feature learning and swarm optimization surpasses single-method approaches.

5. Experiment and performance analysis

This section describes the approach, data analysis, and results of executing the suggested hybrid model incorporating a CNN and an SVM optimized by nature. To comprehend the model's practical efficacy, we examine the dataset, evaluate the model, and analyze the findings using confusion matrix analysis.

Table 2
Overview of datasets utilized.

Datasets	Source(s)	Legitimate	Phishing	Total
D1	UCI repository	1 127	0873	2 000
D2	UCI repository, Phish Tank	3 881	3 110	6 991
D3	Kaggle	6 157	4 898	11 055
D4	Kaggle, UCI repository, Phish Tank, 5000 Best Websites list and Alexa Top Sites	57 000	31 647	88 647

5.1. Dataset

The datasets utilized for training and evaluating the proposed models originate from the UCI Machine Learning Repository [35], Kaggle [36], 5000 Best Websites (<http://5000best.com/websites/>), PhishTank (<http://www.phishtank.com/developer>), and Alexa (<https://www.alexa.com/topsites>). These datasets incorporate a diverse range of web-based features essential for distinguishing phishing websites from legitimate ones. Table 2 provide the datasets exhibit significant variation in size and composition, ensuring a comprehensive evaluation of the model under diverse conditions.

Each dataset was divided into training and testing subsets in an 80:20 ratio, ensuring a comprehensive assessment of the model under varied conditions. The experiments were conducted on Colab Pro, equipped with an NVIDIA A100 Tensor Core GPU and 80 GB of RAM, which provided adequate computational resources for the study.

5.2. Evaluation metrics

To evaluate and optimize the performance of classification models, various metrics derived from the confusion matrix are essential. These metrics provide quantitative insights into the model's classification performance.

Accuracy: Accuracy reflects the proportion of correctly classified instances out of the total predictions, as shown in Eq. (1):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP and TN are the true positives and true negatives, and FP and FN are the false positives and false negatives, respectively.

Precision: Precision measures the accuracy of positive predictions, as described in Eq. (2):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Recall (Sensitivity): Recall evaluates the model's effectiveness in identifying true positive instances among all actual positive cases, as computed in Eq. (3):

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

F1 Score: The F1 Score, expressed in Eq. (4), combines Precision and Recall into a single measure by calculating their harmonic mean:

$$\text{F1Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

False Positive Rate (FPR): The False Positive Rate (FPR), defined in Eq. (5), quantifies the proportion of actual negative instances misclassified as positive:

$$\text{FPR} = \frac{FP}{TN + FP} \quad (5)$$

These metrics, collectively derived from the confusion matrix, provide a robust framework for evaluating classification model performance and identifying areas for optimization.

Table 3
Performance for SVM with nature inspired optimization on dataset 1.

Model	Accuracy	Precision	Recall	F-Score	FPR
SVM Firefly	0.9276	0.9246	0.9497	0.9370	0.1013
SVM BAT	0.9200	0.9476	0.9156	0.9313	0.0736
SVM Whale	0.9200	0.9476	0.9156	0.9313	0.0736
SVM ACO	0.9175	0.9330	0.9207	0.9268	0.0867
SVM GWO	0.9326	0.9313	0.9487	0.9399	0.0876

Table 4
Performance for SVM with Nature Inspired Optimization on Dataset 2.

Model	Accuracy	Precision	Recall	F-Score	FPR
SVM Firefly	0.9276	0.9246	0.9497	0.9370	0.1013
SVM BAT	0.9322	0.9306	0.9487	0.9396	0.0886
SVM Whale	0.9254	0.9222	0.9467	0.9343	0.1019
SVM ACO	0.9335	0.9295	0.9436	0.9365	0.0774
SVM GWO	0.9326	0.9313	0.9487	0.9399	0.0876

Table 5
Performance for SVM with nature inspired optimization on dataset 3.

Model	Accuracy	Precision	Recall	F-Score	FPR
SVM Firefly	0.9249	0.9183	0.9508	0.9343	0.1082
SVM BAT	0.9249	0.9159	0.9518	0.9335	0.1084
SVM Whale	0.9267	0.9155	0.9559	0.9353	0.1094
SVM ACO	0.9299	0.9199	0.9567	0.9379	0.1033
SVM GWO	0.9267	0.9155	0.9559	0.9353	0.1094

Table 6
Performance for SVM with nature inspired optimization on dataset 4.

Model	Accuracy	Precision	Recall	F-Score	FPR
SVM Firefly	0.9199	0.9074	0.9478	0.9264	0.1072
SVM BAT	0.9189	0.9049	0.9489	0.9255	0.1073
SVM Whale	0.9177	0.9066	0.9499	0.9263	0.1084
SVM ACO	0.9179	0.9061	0.9476	0.9269	0.1012
SVM GWO	0.9177	0.9079	0.9487	0.9243	0.1092

5.3. Experiment 1: SVM hyperparameters using nature inspired optimization

The performance of SVM models optimized with five nature-inspired algorithms. The results, presented in Tables 3, 4, 5, and 6, highlight the impact of each optimization method on the SVM models, revealing their respective strengths and weaknesses in terms of accuracy, precision, recall, and false positive rates.

The SVM models optimized with nature-inspired algorithms generally exhibited excellent performance on Dataset 1, as shown in Table 3. The SVM models optimized using the Bat Algorithm (SVM BAT) and Whale Optimization Algorithm (SVM Whale) achieved consistent accuracy and a reduced false positive rate, effectively minimizing misclassifications. While both models performed well, SVM BAT demonstrated superior recall, whereas SVM Whale excelled in precision. The SVM model optimized with Ant Colony Optimization (SVM ACO) delivered balanced results but exhibited a marginally higher false positive rate compared to SVM BAT and SVM Whale. Notably, the SVM model optimized with Grey Wolf Optimizer (SVM GWO) achieved the highest accuracy and maintained an optimal balance between precision and recall, emerging as the most effective model for error reduction and high-quality predictions.

In Dataset 2, illustrated in Table 4, were consistent with those on Dataset 1, albeit with minor variations. The SVM Firefly model, while maintaining reliable accuracy and recall, continued to exhibit a higher false positive rate. The SVM BAT model showed moderate improvements in recall and precision compared to its performance on Dataset 1, though it still faced challenges in reducing false positives. The SVM

Table 7

Performance for CNN and SVM with GWO Optimization.

Model	Accuracy	Precision	Recall	F-Score	FPR
Dataset 1	0.9984	0.9991	0.9984	0.9987	0.0011
Dataset 2	0.9986	0.9992	0.9984	0.9988	0.0010
Dataset 3	0.9995	0.9992	1.0000	0.9996	0.0010
Dataset 4	0.9918	0.9913	0.9941	0.9927	0.0110

Whale model performed slightly worse than SVM BAT on this dataset, with a marginally higher false positive rate. In contrast, the SVM ACO model demonstrated exceptional accuracy, a well-balanced set of metrics, and a relatively low false positive rate, making it a top performer. The SVM GWO model maintained its stability and reliability, consistently delivering strong results across datasets.

The SVM models exhibited consistently robust performance on Dataset 3, as presented in Table 5, despite minor variations across performance metrics. The SVM enhanced by the Firefly Algorithm achieved notably high recall and accuracy; however, it consistently demonstrated an elevated false positive rate, indicating a limitation in achieving balanced classification. The SVM-BAT model-maintained performance levels comparable to those observed on Dataset 2, with slight improvements in recall, yet continued to encounter challenges in mitigating false positive occurrences. The SVM-Whale model, although characterized by a relatively high false positive rate, sustained reliable and accurate classification outcomes. Among the evaluated models, the SVM integrated with the ACO algorithm demonstrated the most balanced performance, achieving an effective trade-off between accuracy, recall, and a controlled false positive rate. While the SVM- GWO model did not surpass the SVM-ACO in overall performance, it exhibited notable resilience, delivering results comparable to those of the SVM-Whale model, thereby underscoring its reliability in diverse classification scenarios.

The performance analysis of SVM models optimized with nature-inspired techniques on Dataset 4, presented in Table 6. The Firefly algorithm slightly outperformed others in overall accuracy and balance, while the ACO algorithm distinguished itself with a lower false positive rate, underscoring its reliability in minimizing classification errors. The differences between the models were minimal, suggesting that all nature-inspired optimization techniques effectively enhance SVM performance, making them suitable for robust applications such as phishing detection.

5.4. Experiment 2: CNN-based feature extraction with GWO-optimized SVM

Table 7 provide the effectiveness of the hybrid model, that combines CNN and SVM and incorporates GWO optimization, in identifying phishing web pages. The model exhibits high levels of accuracy, precision, and recall across diverse data sets, exhibiting minimal false detections and successfully detecting almost all genuine phishing threats. The model performs best within an acceptable standard and effectively distinguishes between legitimate and malicious websites, its critical role in cyberspace defense strategies is undeniable. Due to its low false positive rates, this approach proves useful in practice as it can be easily integrated into the overall security framework to mitigate phishing threats.

Statistical Analysis: The hybrid CNN+SVM model, integrated with GWO-based optimization, demonstrates exceptional reliability and consistency, as evidenced by its performance across 30 independent evaluations on multiple datasets in Table 8. The model achieves consistently high metrics, with minimal standard deviations for accuracy, precision, recall, and F1-score, showcasing its robustness and repeatability. For instance, Dataset 1 achieved an average accuracy of 0.9982 with a standard deviation of 0.0001, while Dataset 3 excelled with an accuracy of 0.9995 and a standard deviation of 0.0002. This consistent performance across datasets highlights the model's ability to generalize effectively to unseen data, making it a practical and robust tool for combating phishing threats in real-world cybersecurity scenarios.

Analysis of Variance (ANOVA): In this study, we conducted a comprehensive evaluation of various machine learning approaches for binary classification, with particular focus on our proposed hybrid CNN+SVM-GWO model. Statistical analysis reveals significant performance differences among the classification methods tested (ANOVA: $F=9.18$, $p<0.001$). Among standard machine learning approaches, SVM and Logistic Regression (both 92.65 % with standard deviations of 0.36 % and 0.25 %, respectively), while K-Nearest Neighbors demonstrated inferior performance ($91.59\% \pm 0.37\%$). Pairwise t-tests confirmed that KNN performed significantly worse than all other standard models ($p<0.01$ for all comparisons), SVM, and Logistic Regression were not statistically significant ($p>0.05$). The hybrid CNN+SVM-GWO model substantially outperformed all conventional approaches, achieving an exceptional accuracy of 99.18 %. The extremely high z-score (12.38) when comparing against the distribution of standard model accuracies indicates that this performance improvement is highly statistically significant ($p\ll 0.0001$). The optimized SVM hyperparameters ($C=6.60$, $\gamma=1.32$) determined by the Grey Wolf Optimizer algorithm proved highly effective when applied to CNN-extracted features. These metrics collectively suggest that the combination of deep feature extraction through CNN and metaheuristic-optimized SVM classification represents a superior approach for this classification task, with performance improvements that are both substantial and statistically validated.

Confusion Matrix Analysis: The performance of the CNN + SVM-GWO model in classifying phishing and legitimate websites is illustrated in Figs. 3, 4, 5 and 6 through their associated confusion matrix analysis about datasets 1, 2, 3, and 4, respectively.

In Dataset 1, the model correctly detects the majority of cases, with only a few cases missing and misclassified, demonstrating its robustness. In Dataset 2, the model demonstrates strength once more, correctly anticipating most cases and only committing errors in a small number of cases, thereby reinforcing its reliability. Dataset 3 displays nearly flawless classification, with no false alarms and only one false alarm reported, underscoring the model's comprehension of the assigned tasks, as it records minimal errors during classification. In dataset 4, the confusion matrix demonstrates the effectiveness of the CNN+SVM-GWO model in phishing detection, with 6,211 true positives and 7,857 true negatives, alongside minimal errors are 69 false negatives and 47 false positives. This indicates a high level of accuracy and balanced detection, as the model successfully identifies phishing and legitimate websites with very low misclassification rates. The strong performance suggests that the combination of CNN for feature extraction and SVM-GWO

Table 8

Mean and standard deviation for CNN and SVM with GWO optimization.

References	Accuracy		Precision		Recall		F1-Score	
	Mean	Std Dev	Mean	Std Dev	Mean	Std Dev	Mean	Std Dev
Dataset 1	0.9982	0.0001	0.9991	0.0001	0.9984	0.0001	0.9987	0.0002
Dataset 2	0.9986	0.0003	0.9992	0.0003	0.9984	0.0004	0.9988	0.0002
Dataset 3	0.9995	0.0002	0.9992	0.0002	1.0000	0.0001	0.9996	0.0003
Dataset 4	0.9918	0.0004	0.9913	0.0004	0.9941	0.0003	0.9927	0.0004

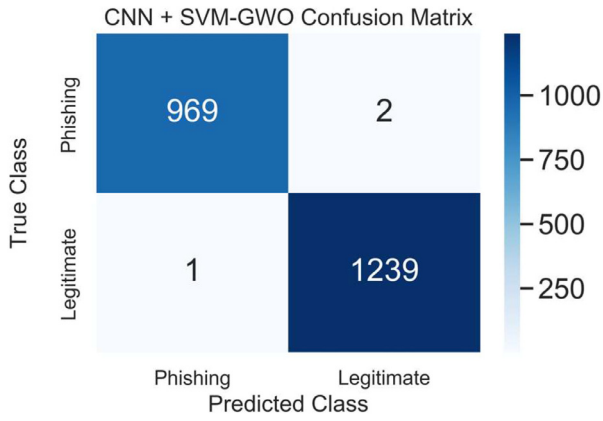


Fig. 3. Dataset 1.

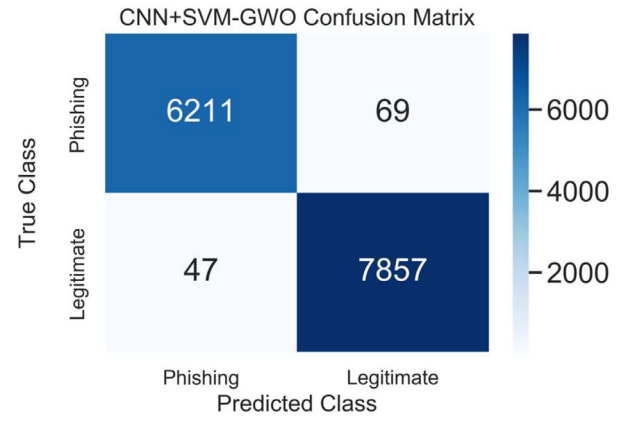


Fig. 6. Dataset 4.

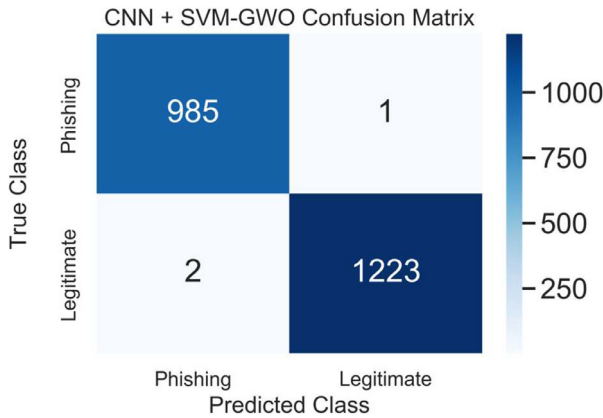


Fig. 4. Dataset 2.

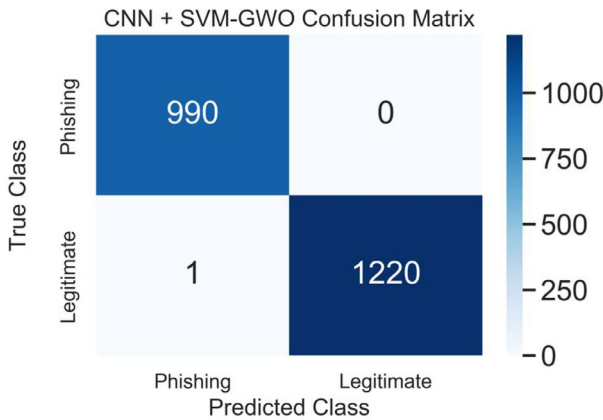


Fig. 5. Dataset 3.

for optimization and classification is robust and capable of generalizing well across different samples.

5.5. Experiment 3: Feature extraction using CNN-LSTM ensemble

In Fig. 7, a CNN and LSTM ensemble model is employed to detect phishing websites by harnessing both spatial and temporal features within the data. The proposed system is evaluated using Dataset 4, ensuring comprehensive coverage of phishing scenarios. The spatial feature extraction stage utilizes two dimensional convolutional layers with 32 filters (1×1 kernels) and ReLU activation, followed by a 1×1 max pooling operation to reduce dimensionality. This process

Table 9

Performance Comparison for Ensemble Methods.

Model	Accuracy	Precision	Recall	F-Score	FPR
CNN-LSTM-SVM	0.9854	0.9831	0.9911	0.9870	0.0218
CNN-LSTM+SVM-GWO	0.9889	0.9863	0.9941	0.9902	0.0177

is repeated to reinforce feature abstraction. The resulting feature maps are flattened into a vector and reshaped into sequential input for the LSTM components. A two-layer LSTM architecture, comprising a 128-unit LSTM followed by a 64-unit LSTM, captures temporal dependencies. The model integrates hyperparameter optimization of a support vector machine through the grey wolf optimizer, which refines key parameters such as the SVM kernel and regularization coefficient. This approach bolsters classification performance. A fully connected layer classifies websites as either phishing or legitimate. The combination of convolutional feature extraction, sequential modeling, and advanced optimization demonstrates increased robustness and adaptability to evolving phishing tactics.

The comparison of ensemble methods highlights differences in classification performance for phishing detection in Table 9. The CNN-LSTM+SVM-GWO model shows improvements across all metrics. The integration of the GWO allows better feature selection and parameter tuning, leading to superior classification results. Accuracy, precision, recall, and F-score reach higher levels, while FPR decreases, reducing incorrect phishing classifications. The inclusion of optimization techniques such as GWO enhances model effectiveness. The CNN-LSTM+SVM-GWO model demonstrates a more refined balance between accuracy and false positives, ensuring better phishing detection performance.

The confusion matrices for the CNN-LSTM-SVM method are illustrated in Fig. 8, while the confusion matrices for the CNN-LSTM+SVM-GWO model are presented in Fig. 9. The optimized model demonstrates improvements in phishing detection by reducing misclassifications, enhancing recall, and lowering the false positive rate. A higher number of phishing and legitimate websites are correctly classified, reflecting better accuracy and precision. The integration of the GWO enhances feature selection and parameter tuning, leading to a more refined balance between detection capability and classification errors. The CNN-LSTM+SVM-GWO model exhibits superior performance, making it a more effective approach for phishing website detection.

5.6. Experiment 4: Transformer model for phishing detection

This experiment compares two phishing URL detection models, namely a transformer-based model and a bidirectional encoder representation from transformers (BERT) based model, each designed to classify URLs as phishing or legitimate. The Transformer based model utilizes a transformer encoder block that incorporates layer normalization to sta-

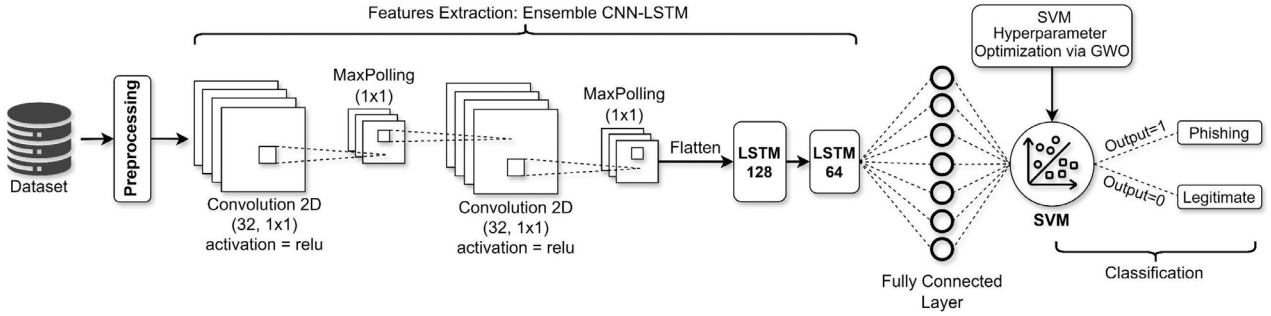


Fig. 7. Feature extraction using ensemble CNN-LSTM+SVM-GWO.

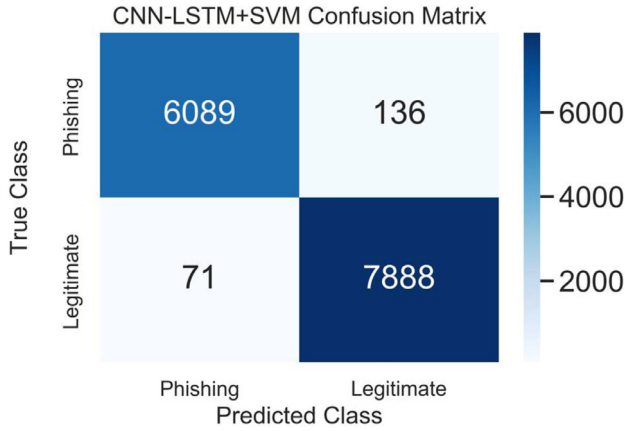


Fig. 8. CNN-LSTM-SVM.

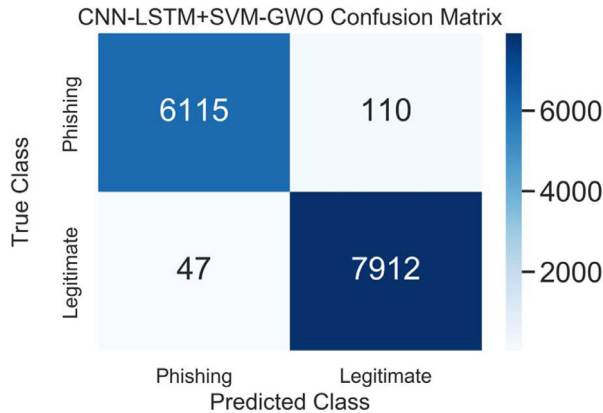


Fig. 9. CNN-LSTM+SVM-GWO.

bilize training, multi head self-attention to capture long range dependencies, and residual connections to address the vanishing gradient issue. It also includes a feed forward network with two Conv1D layers, the first of which uses ReLU activation, allowing for nonlinear representation learning while maintaining the original input size. The BERT based model employs a pre trained Bert base uncased model and its tokenizer to produce input IDs and attention masks of fixed length (128 tokens), and it is trained for three epochs with early stopping and learning rate reduction callbacks to prevent overfitting. As shown in Table 10, the performance analysis reveals that the transformer model outperforms BERT in terms of classification accuracy, precision, recall, and F score, indicating a higher capacity to distinguish phishing URLs from legitimate ones while effectively balancing the minimization of false positives and the identification of malicious instances.

Table 10

Performance comparison for ensemble methods.

Model	Accuracy	Precision	Recall	F-Score
Bert	0.8619	0.8622	0.8619	0.8619
Transformer	0.9101	0.9112	0.9101	0.9100

5.7. Internal performance comparison

The comparison of different phishing detection models reveals distinct variations in performance as shown in Table 11. Traditional SVM-based models provide reliable results but struggle with balancing precision and recall, often resulting in false positives. The transformer model offers an improvement, leveraging self-attention mechanisms to capture long-range dependencies in the data, allowing for better classification. Further advancements are seen in ensemble models that integrate CNN and LSTM, which enhance feature extraction and sequential learning, improving overall performance. The CNN-SVM+GWO proposed model emerges as the best performer, combining CNN for feature extraction with SVM fine-tuned through GWO, achieving near-perfect accuracy and classification.

5.8. Comparison with current state-of-the-art methods

To evaluate the efficacy of proposed phishing website detection model, we conducted a comprehensive comparative analysis against existing state-of-the-art methods across three widely used datasets: Kaggle, the UCI Machine Learning Repository, and a Combined Dataset integrating multiple real-world sources. Tables 12–14 summarize the performance of various algorithms reported in recent literature and benchmark them against our proposed model, which utilizes a hybrid CNN + SVM architecture with hyperparameter optimization using GWO.

5.8.1. Performance on Kaggle dataset

As illustrated in Table 12, our proposed method significantly outperforms existing techniques applied to the Kaggle phishing dataset. The highest accuracy reported among previous studies was 99.04 % by Zhu et al. (2022) using traditional machine learning techniques. In contrast, our model achieves a superior accuracy of 99.95 %, indicating that the integration of deep learning with evolutionary optimization offers a considerable edge in generalization and prediction capabilities. Other noteworthy approaches such as deep learning by Kumar et al. (2023) and metaheuristic-enhanced RNN by Gupta et al. (2024) reached 94.80 % and 92 %, respectively, highlighting a clear gap bridged by our solution.

5.8.2. Performance on UCI machine learning repository dataset

In Table 13, our model again demonstrates its superiority with an accuracy of 99.84 %, surpassing all listed contemporary methods. Notable competitors include fuzzy modeling by Nordin et al. (2023) with 98.80 % accuracy and the rabbit optimization + ensemble method by Shahba

Table 11
Performance comparison for machine and deep learning models.

Method	Accuracy (%)	Precision (%)	Recall (%)	F-Score (%)
SVM Firefly	91.99	90.74	94.78	92.64
SVM BAT	91.89	90.49	94.89	92.55
SVM Whale	91.77	90.66	94.99	92.63
SVM ACO	91.79	90.61	94.76	92.69
SVM GWO	91.77	90.79	94.87	92.43
Bert	86.19	86.22	86.19	86.19
Transformer	91.01	91.12	91.01	91
CNN-LSTM-SVM	98.54	98.31	99.11	98.7
CNN-LSTM+SVM-GWO	98.89	98.63	99.41	99.02
CNN-SVM+GWO	99.18	99.13	99.41	99.27

Table 12
Phishing website detection methods using Kaggle.

Reference	Algorithm	Accuracy (%)
Saha et al. (2020) [37]	Random Forest, Decision Tree	93
Zhu et al. (2022) [27]	Machine learning	99.04
Kumar et al. (2023) [29]	Deep learning	94.80
Gupta, B. B. et al. (2024) [38]	Whale Optimization Algorithm + RNN	92
Our Proposed Work	CNN + SVM Hyperparameter Optimization using GWO	99.95

Table 13
Phishing website detection methods using UCI machine learning repository.

Reference	Algorithm	Accuracy (%)
Basit et al., (2020) [39]	K-Nearest Neighbor, Random Forest, Decision Tree, Artificial Neural Network	97.33
Ali, W. et al. (2020) [40]	PSO-based Feature Weighting	92.19
Nordin et al. (2023) [28]	Fuzzy Modelling	98.80
Shahba et al. (2025) [41]	Rabbit Optimization Algorithm + Voting Ensemble	97.91
Our Proposed Work	CNN + SVM Hyperparameter Optimization using GWO	99.84

Table 14
Comparison of phishing website detection methods using combined dataset.

Reference	Algorithm	Data Set	Accuracy (%)
Ali, Waleed et al. (2019) [42]	GA + Deep Neural Networks (DNNs)	UCI Repository and phishtank	88.77
Zhu et al. (2022) [27]	Machine learning	Kaggle, Mendeley Data website	99.04
Anupam and Kar, (2021) [26]	SVM, Bat Algorithm, Whale Optimization, Grey Wolf Optimizer, Firefly Algorithm	PhishTank, Yahoo, UCI machine learning repository	90.38
Kocyigit et al. (2024) [43]	GA-based Feature Selection	High-Risk URL and Content Dataset	91.78
Our Proposed Work	CNN + SVM Hyperparameter Optimization using GWO	Kaggle, UCI machine learning repository, Phishtank, 5000 Best Websites and Alex)	99.18

et al. (2025) at 97.91 %. Basit et al. (2020) achieved a relatively high accuracy of 97.33 % using traditional ensemble methods, they lag behind the hybrid optimization strategy employed in our work. The significant margin of improvement highlights the effectiveness of combining CNN feature extraction with SVM classification and fine-tuning via GWO.

5.8.3. Performance on combined dataset

Evaluation on a combined dataset, as seen in Table 14, presents a more rigorous test by incorporating diverse data sources, including Phishtank, UCI, Kaggle, and others. Once again, our proposed framework achieves the highest accuracy of 99.18 %, outperforming Zhu et al. (2022) at 99.04 % and significantly ahead of methods like genetic algorithm-based deep networks by Ali et al. (2019) and multi-algorithm optimizers by Anupam and Kar (2021), which achieved 88.77 % and 90.38 %, respectively. This demonstrates the scalability and adaptability of our approach across heterogeneous and complex datasets.

Across all datasets, our CNN + SVM model optimized with GWO consistently surpasses traditional machine learning, standalone deep learning, and metaheuristic hybrid models. The superior accuracy highlights its potential as a robust and scalable solution for phishing website detection in real-world cybersecurity applications.

6. Discussions and limitations

The proposed method framework for phishing website detection demonstrates improved performance compared to existing approaches, achieving an accuracy of 99.18 %. This level of accuracy is higher than those typically reported for conventional machine learning techniques, including k-nearest neighbor, random forest, and logistic regression, which generally reach around 97 % [37,39,42], as well as more recent hybrid optimization and deep learning-based methods such as fuzzy modeling (98.80 %) [28] and deep learning techniques (94.80 %) [29]. The use of a convolutional neural network enables the extraction of complex, hierarchical features from phishing datasets, allowing the model to identify patterns that may not be captured by traditional approaches. The support vector machine, with hyperparameters optimized through the grey wolf optimizer, contributes to forming clear decision boundaries for distinguishing between phishing and legitimate websites. The optimization process also removes the need for manual tuning, enhancing the model's consistency and scalability. Future research could explore the integration of multi-modal data, such as textual and visual elements from web pages, to improve adaptability across a broader range of phishing scenarios. These aspects of-

fer meaningful directions for further work in phishing detection and cybersecurity.

7. Conclusion and future scope

The proposed hybrid framework integrates a convolutional neural network for deep feature extraction with a support vector machine optimized through the grey wolf optimizer, has shown outperformance in phishing website detection. The convolutional neural network effectively captured the intricate, nonlinear patterns inherent in phishing websites, while the optimized support vector machine established a refined classification boundary that minimized both false positives and false negatives. This synergy between deep feature extraction, precise classification, and nature inspired optimization set a new benchmark in phishing detection, achieving a 99.18 % accuracy with minimal false positives. The findings illustrate the transformative potential of combining deep learning, machine learning, and nature inspired optimization for addressing cybersecurity threats. Future research can investigate incremental and online learning methods for adapting to rapidly evolving attacks, employ advanced model explainability approaches to enhance trust and transparency, and incorporate additional features to further improve robustness.

Data availability

Not Applicable.

Ethical approval

This article does not contain any studies involving human participants or animals conducted by any of the authors.

Funding

This work was not funded by any organization.

Declaration of competing interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Santosh Kumar Birthriya: Writing – original draft, Methodology, Conceptualization. **Priyanka Ahlawat:** Writing – review & editing, Supervision. **Ankit Kumar Jain:** Writing – review & editing, Supervision.

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